Mukesh Singh

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Research Interests

My passion lies in understanding the fundamental properties of nanomaterials, including their electronic, optical, anharmonic, and elastic characteristics, and predicting novel materials using ab initio methods and machine learning approaches. These attributes of material extend my interest in practical applications, such as thermoelectric properties, thermal conductivity, batteries (metal-ion and metal-Sulfur), catalysis reactions like Hydrogen Evolution Reactions and Oxygen Evolution Reactions etc, and hydrogen storage.

Research Experience

Central Michigan University — Mount Pleasant, USA
Postdoctoral research fellow — Jan, 2025 – present
First-principles study of electrode materials for metal–sulfur batteries.

Indian Institute of Technology (IIT) Bombay — Mumbai, India
Postdoctoral research fellow — Apr 2024 – Dec 2024
Quantum dot modeling and 2D materials: electronic, optical, and thermoelectric properties.

Education

Indian Institute of Technology (IIT) Bombay — Mumbai, India
Ph.D. in Computational Condensed Matter Physics — 2019 – 2024
Thesis: Structural and Electronic Properties of 2D Materials and their Applications in Hydrogen Production and Hydrogen Storage: First Principles Study

Kirori Mal College, University of Delhi — Delhi, India
M.Sc. in Physics — 2015 – 2017
Thesis: Stringy Correction to Reissner-Nordström Black Hole: Multi-Universe Scenario

Deen Dayal Upadhyay Gorakhpur University — Gorakhpur, India
B.Sc. in Physics, Chemistry, Mathematics — 2012 – 2015

Technical Skills

Projects: Vestapy, RxnPy, VaspWorks, Pre_and_post_processing_dft (see github.com/mukesh4iitb).
Programming: Python, Fortran, Bash (awk, sed), C; C++ (basic), machine learning.
Simulation & DFT: VASP, Quantum ESPRESSO, Gaussian, AFLOW.
Materials Modeling: ASE, Pymatgen.
Analysis Tools: Phonopy, Phono3py, ShengBTE/AlmaBTE, BoltzTrap2.
Software: LaTeX, LibreOffice, VESTA, XCrysDen, GIMP, Inkscape, Origin-lab.

Awards & Service

Awards: CSIR-SRF (2021); CSIR-JRF (2017, 2018); INSPIRE Fellow (Top 1%, 2012); NGPE: Top 1% (2015), Top 10% (2013, 2014).

Positions of Responsibility: Academic Unit Representative (AURAA, 2020–2022); General Secretary, RSAP (2020–2021); System Administrator, H14 IIT Bombay (2022–2023).
Teaching Assistant: Teaching Assistant for courses in Python, Fortran simulations, E&M, and Thermal Physics.

References

Prof. Alok Shukla
(Ph.D. Supervisor)
Professor
IIT Bombay, India
Department of Physics
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Prof. Aftab Alam
Professor
IIT Bombay, India
Department of Physics
email: aftab@phy.iitb.ac.in

Prof. Veronica Barone
(Postdoctoral Supervisor)
Professor
Central Michigan Univeristy, USA
Department of Physics
email: baron1v@cmich.edu

Publications

[1] Jyoti Pandey, Aliakbar Yazdani, **Mukesh Singh**, .., and Bradley D. Fahlman. “Regulating Polysulfide Conversion on Fe–N_x Single-Atom Catalysts Supported on Graphitic Carbon Nitride Revealed by Semi-In Situ XPS of Li-S Batteries”. (Submitted). 2026.

[2] **Mukesh Singh** and Veronica Barone. “Unified Thermodynamic and Kinetic Criteria for Screening 2D Anchoring Materials in Li–S, Na–S, and K–S Batteries”. (Accepted in ACS Applied Energy Materials). 2026.

[3] **Mukesh Singh**, Surinder Pal Kaur, and Brahmananda Chakraborty. “Modeling and tuning the electronic, mechanical and optical properties of a recently synthesized 2D polyaramid: a first principles study”. In: *Phys. Chem. Chem. Phys* 26 (July 2024), p. 21874. doi: 10.1039/D4CP02027H. url: https://doi.org/10.1039/D4CP02027H.

[4] **Mukesh Singh**, Alok Shukla, and Brahmananda Chakraborty. “Improving hydrogen evolution catalytic activity of 2D carbon allotrope biphenylene with B, N, P doping: Density functional theory investigations”. In: *International Journal of Hydrogen Energy* (2024), pp. 569–579. doi: https://doi.org/10.1016/j.ijhydene.2023.08.359.

[5] Pratap Mane, Surinder Pal Kaur, **Mukesh Singh**, Ajit Kundu, and Brahmananda Chakraborty. “Superior hydrogen storage capacity of Vanadium decorated biphenylene (Bi+V): A DFT study”. In: *International Journal of Hydrogen Energy* 48 (2023), pp. 28076–28090. doi: https://doi.org/10.1016/j.ijhydene.2023.04.033.

[6] **Mukesh Singh** and Brahmananda Chakraborty. “2D BN -biphenylene: structure stability and properties tenability from a DFT perspective”. In: *Phys. Chem. Chem. Phys.* 25 (2023), pp. 16018–16029. doi: 10.1039/D3CP00776F.

[7] **Mukesh Singh**, Alok Shukla, and Brahmananda Chakraborty. “High capacity hydrogen storage on zirconium decorated γ-graphyne: A systematic first-principles study”. In: *International Journal of Hydrogen Energy* 48 (2023), pp. 37834–37846. doi: https://doi.org/10.1016/j.ijhydene.2022.07.062.

[8] **Mukesh Singh**, Alok Shukla, and Brahmananda Chakraborty. “Highly efficient hydrogen storage of a Sc decorated biphenylene monolayer near ambient temperature: ab initio simulations”. In: *Sustainable Energy Fuels* 7 (2023), pp. 996–1010. doi: 10.1039/D2SE01351G.

[9] **Mukesh Singh**, Alok Shukla, and Brahmananda Chakraborty. “An Ab-initio study of the Y decorated 2D holey graphyne for hydrogen storage application”. In: *Nanotechnology* 33 (July 2022), p. 405406. doi: 10.1088/1361-6528/ac7cf6. url: https://dx.doi.org/10.1088/1361-6528/ac7cf6.